Q6 NO CALCULATOR ALLOWED Q6

Answer QUESTION 6 PARTS A and B on this page.

PART A

$$\sum_{n=1}^{\infty} \frac{(x-4)^{n+1}}{(n+1)3^n} \qquad \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{(x-4)^{n+2}}{(n+2)3^{n+1}}\cdot\frac{3^n(n+1)}{(x-4)^{n+1}}\right| = \lim_{n\to\infty}\left|\frac{x-4}{3}\,\frac{(n+1)}{(n+2)}\right|$$

$$\left|\frac{x-4}{3}\right| < 1$$

$$-1 < \frac{x-4}{3} < 1$$

$$-3 < x-4 < 3$$

$$\boxed{1 \le x < 7}$$

$x=1$: $\displaystyle\sum_{n=1}^{\infty} \frac{(-3)^{n+1}}{(n+1)3^n} = \frac{(-1)^{n+1}\cdot 3}{n+1}$

$\displaystyle\lim_{n\to\infty}\frac{\frac{3}{n+1}}{\frac{3}{n}} = 1 \quad 0<1<\infty$ diverges by LCT to $\displaystyle\sum_{n=1}^{\infty}\frac{3}{n}$

AST: $\displaystyle\lim_{n\to\infty}\frac{3}{n+1} = 0$ ✓, $\displaystyle\frac{3}{n+1} > \frac{3}{n+2}$ ✓

Converges conditionally by AST

$x=7$: $\displaystyle\sum_{n=1}^{\infty}\frac{3^{n+1}}{(n+1)3^n} = \sum_{n=1}^{\infty}\frac{3}{n+1}$

Diverges by LCT

PART B

$$f = \frac{(x-4)^2}{6} + \frac{(x-4)^3}{27} + \frac{(x-4)^4}{4\cdot 27}$$

$$f' = \frac{2(x-4)}{6} + \frac{3(x-4)^2}{27} + \frac{4(x-4)^3}{4\cdot 27} + \dots$$

$$f' = \frac{x-4}{3} + \frac{(x-4)^2}{3^2} + \frac{(x-4)^3}{3^3} + \dots \frac{(x-4)^n}{3^n}$$

$$\sum_{n=1}^{\infty}\frac{(x-4)^n}{3^n}$$

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Answer QUESTION 6 PARTS C and D on this page.

PART C

Geometric series $\sum_{n=1}^{\infty} \frac{(x-4)^n}{3^n}$ $|r|<1$ and the series converges to $\frac{a}{1-r}$ in the interval of convergence.

$r = \frac{x-4}{3}$, and $a = a_1 = \frac{x-4}{3}$.

$$\frac{a}{1-r} = \frac{\frac{x-4}{3}}{1-\frac{x-4}{3}} = \frac{\frac{x-4}{3}}{\frac{3-x+4}{3}} = \frac{x-4}{7-x}$$

PART D

No. The interval of convergence for f about $x=4$ is $1 \le x < 7$, and $x=8$ is outside of that range. Since the radius of convergence is the same for f' and f, and both are centered about $x=4$, 8 must be out of the interval of convergence for f' as well. Therefore, it must diverge and cannot converge to $f'(x) = \frac{x-4}{7-x}$ at $x=8$.

check: geometric, $|r|<1$ $-1 < \frac{x-4}{3} < 1$

$-1 < x < 7$

geometric, so not

$-1 \le x < 7$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.